

Day 8 — The Doṣa Lens: Heating and Cooling

Module: Auśadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can sort all six module herbs into heating (*uṣṇa*) and cooling (*śīta*) categories, explain the Ayurvedic concept of *vīrya* in one sentence, link this classification to specific *doṣa* imbalances from Module 4, and identify one example where the traditional classification matches a measurable modern pharmacological effect.

Materials

- A printed 6-card set with each herb (one per pair); these can be drafts of the project deck
- Whiteboard with two columns: **uṣṇa** (heating) | **śīta** (cooling)
- Sticky notes
- Optional: Day 4–7 worksheets students completed, so they have their own data

Vocabulary introduced today

- *vīrya* (IAST: *vīrya*; lit. “potency, energy”) — in Ayurveda, the heating or cooling *potency* of a substance, distinct from its taste (*rasa*) and its post-digestive effect (*vipāka*). At Middle band we use only the heating/cooling axis.
- *uṣṇa* (IAST: *uṣṇa*; lit. “hot, heating”) — the heating *vīrya*.
- *śīta* (IAST: *śīta*; lit. “cold, cooling”) — the cooling *vīrya*.

Lesson flow

Warm-up (5 min)

Daily prompt: “Which herb did you smell or eat at home since our last class?” Two students share for 30 sec each.

Now the day’s hook: “From what you remember about each herb — which feels ‘warming’ when you eat it? Which feels ‘cooling’?” Take a few intuitions, write them on the board.

Core (20 min)

1. **What is *vīrya*?** Write on board:

vīrya — the **heating** or **cooling** potency of a substance, measured by its effect on the body.

This is one of four properties Ayurveda uses to classify any food or herb:

- *rasa* — taste (sweet, sour, salty, pungent, bitter, astringent)
- *vīrya* — heating or cooling potency
- *vipāka* — post-digestive effect

- *prabhāva* — special, idiosyncratic action

At Middle band, we focus on *vīrya*. The full fourfold framework comes back at Senior band.

2. **Sort the six herbs.** Build the two-column table on the board, asking students to predict before you confirm:

<i>uṣṇa</i> (heating)	<i>śīta</i> (cooling)
<i>tulasī</i> — eugenol, warming aromatic	<i>pudīnā</i> — menthol, TRPM8 cooling
<i>haridrā</i> — turmeric, mild warming	
<i>adraka</i> — ginger, strongly warming (TRPV1)	
<i>yavānī</i> — ajwain, warming digestive	
<i>nimba</i> — neem (slightly contested — see below)	

The contested case: *nimba* (neem) is classified as bitter (*tikta*) and *śīta* (cooling) in classical Ayurveda — even though it’s a bitter herb without an obviously cooling sensation. Why? Because in the *doṣa* model, bitterness reduces *pitta* (the “fire” *doṣa*) — and a substance that reduces *pitta* is by definition cooling in the system. Bring this up explicitly to show students that the Ayurvedic categories are **functional**, not always sensory.

3. **Why does heating/cooling matter?** Link back to *doṣas* (refresh briefly if Module 4 hasn’t been taught):
 - **Vāta** (motion, air-and-space) — generally treated with warming, moistening, grounding foods/herbs. Cold and dry aggravate *vāta*.
 - **Pitta** (transformation, fire-and-water) — treated with cooling, mild foods. Heat and pungency aggravate *pitta*.
 - **Kapha** (structure, water-and-earth) — treated with warming, light, drying foods. Cold and heavy aggravate *kapha*.

So:

- **Ginger + ajwain** → useful for *kapha* and *vāta* imbalances (warming, stimulating).
- **Mint** → useful for *pitta* imbalances (cooling).
- **Tulsi** → balanced for *vāta/kapha* (warming, but not as strongly as ginger).
- **Turmeric** → mildly heating but broadly used because of its antimicrobial profile.
- **Neem** → useful for *pitta* and *kapha* imbalances (its bitterness reduces both, in the framework’s logic).

4. **The match between tradition and modern pharmacology.** This is the most important slide of the module:

Herb	Tradition says	Modern says
<i>pudīnā</i> (mint)	<i>śīta</i> — cooling	Menthol activates TRPM8 → “cool” signal to brain
<i>adraka</i> (ginger)	<i>uṣṇa</i> — heating	Gingerol activates TRPV1 → “warm” signal to brain

Both classifications were made by careful observers — millennia apart, using completely different methods. They agree because they were both tracking a real biological effect.

5. **Source — paraphrase of the Auṣadhi-sūkta** (Ṛgveda 10.97). Read aloud once:

“O herbs, born before the gods, brown and bright, born of past ages — may you who carry many forms protect the one who calls on you.”

— paraphrase of the opening of Ṛgveda 10.97, the *Auṣadhi-sūkta*

Comment: the hymn is **3000+ years old**. The plants are addressed as agents — like persons. Whether or not you take that literally, notice what it does: it treats the plant as something to **pay attention to**. The *doṣa* framework is built on millennia of paying that kind of attention.

Activity (15 min) — Sort by *vīrya* (see [activities/activity-02-sort-by-virya.md](#))

Pairs receive a set of 12 cards: the 6 module herbs + 6 other common kitchen items (rice, ghee, lemon, ice cream, chilli, coriander seed). They sort into *uṣṇa* / *śīta* / *not sure* and defend their reasoning.

Last 3 min: each pair shares their **most uncertain** card and why.

Wrap-up (5 min)

- Restate safety rule.
- Hand out the project deck instructions for Day 9. Show the rubric on the board. Walk through one card’s expected content.
- Preview Day 9: *“Tomorrow you build your six-card herb deck. Bring your sketches, your notes, your colour pencils.”*

Student-facing content

Vīrya is the heating or cooling *potency* of a substance in Ayurveda.

Heating (*uṣṇa*) — tulsi, turmeric, ginger, ajwain. **Cooling (*śīta*)** — mint. Neem too — **though in a functional, not sensory, way.**

Why does this matter? - Heating herbs balance *vāta* and *kapha* imbalances (cold-related, sluggish conditions). - Cooling herbs balance *pitta* imbalances (heat-related, inflammatory conditions).

Two beautiful matches between tradition and modern science: - *Pudīnā* is *śīta* — menthol activates TRPM8, the same skin receptor that fires when you actually feel

cold. - *Adraka* is *uṣṇa* — gingerol activates TRPV1, the same skin receptor that fires when you actually feel heat.

Both classifications were made by careful observers, millennia apart, using completely different methods. They agree because they were tracking a real biological effect.

“O herbs, born before the gods, brown and bright, born of past ages — may you who carry many forms protect the one who calls on you.” — paraphrase of the opening of the *Auṣadhi-sūkta* (Ṛgveda 10.97)

Homework

- Finalise your project pair. Decide which 2 herbs you’ll each draft cards for first (you’ll do 3 each). Bring everything tomorrow.
- One sentence answer for the next class: “*What is the most surprising thing you’ve learned about herbs in the past week?*”

Differentiation

- **One tier up:** The full Ayurvedic fourfold framework is *rasa-vīrya-vipāka-prabhāva*. Look up what *vipāka* means and find one example where a herb’s *vipāka* differs from its *rasa*.
- **One tier down:** Focus on the heating/cooling sort. The *doṣa* connection comes back in summative review.

Teacher notes

- The TRPM8/TRPV1 ↔ *śīta/uṣṇa* convergence is the cleanest “tradition and modern science describe the same biological reality” moment in the module. Lean into it. Don’t oversell (“Ayurveda discovered TRP channels first” — no, they didn’t, that’s a different question) — but don’t undersell either.
- The neem-is-cooling-because-bitter point will surprise students. Use it to discuss what “potency” means in a functional framework vs a sensory one.
- The *Auṣadhi-sūkta* paraphrase is *not* a translation. Don’t pass it off as a quote. Be explicit: this is a paraphrase of the opening of Ṛgveda 10.97. Teachers who want a published translation should consult Griffith (1896) or Jamison & Brereton (2014).

Citation(s) used in this lesson

- *Auṣadhi-sūkta* — Ṛgveda 10.97 — referenced and paraphrased. Public-domain Vedic source.
- *Caraka Saṁhitā* — *vīrya* as one of four properties of substances. Chapter-level reference to *Sūtrasthāna*.
- TRPM8 / TRPV1 work — established neuroscience. 2021 Nobel Prize in Physiology or Medicine to David Julius and Ardem Patapoutian.